

Processus de rayonnement et photométrie

M1 Astrophysique – CCP HAP702P Astrophysique

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Outline

1 Radiation processes

2 Astronomical photometry

3 Bibliographic ressources

Outline

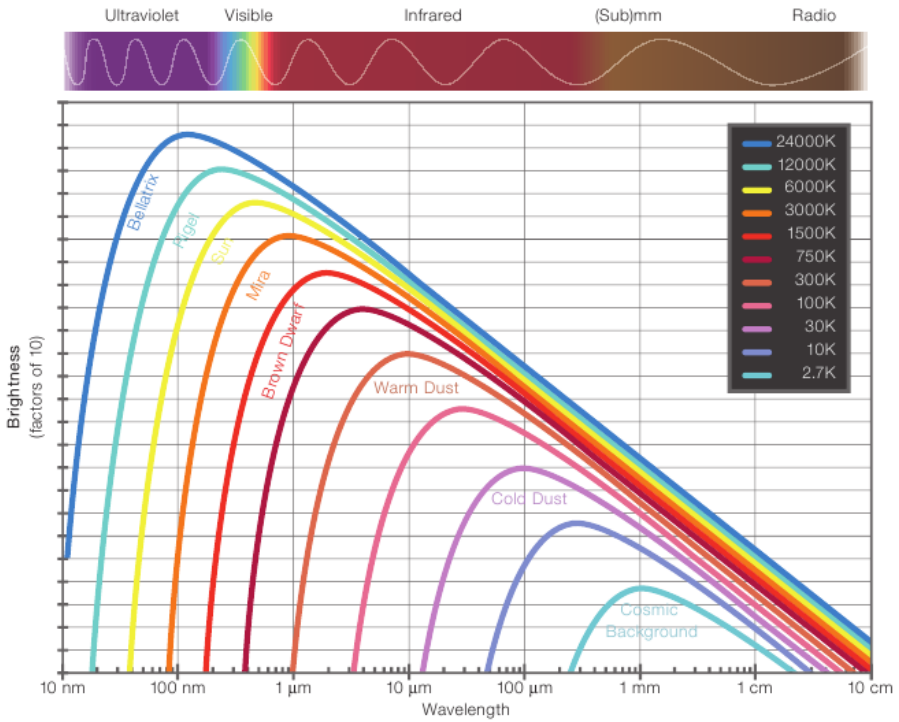
1 Radiation processes
- Various radiation processes
- Properties of the Black-Body Radiation
- An example: the Milky-Way

2 Astronomical photometry

3 Bibliographic ressources

Various radiation processes

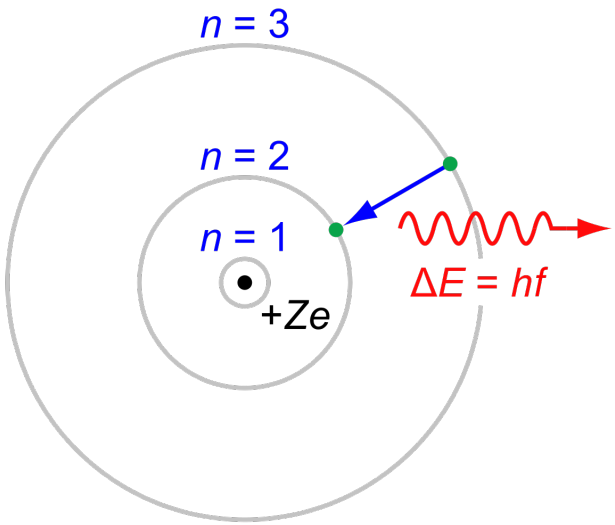
- black-body radiation



Blackbody SED as a function of temperature
Credit : Robert Hurt/ESO

Various radiation processes

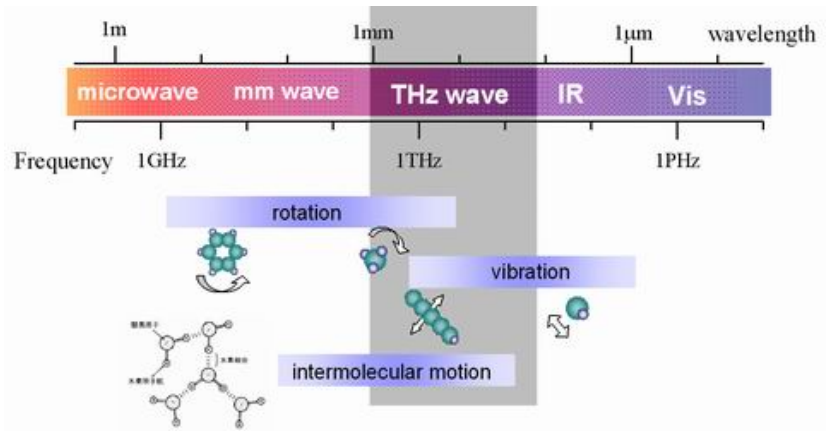
- black-body radiation
- bound-bound
  - electron transition atomic level



Photon emission

Various radiation processes

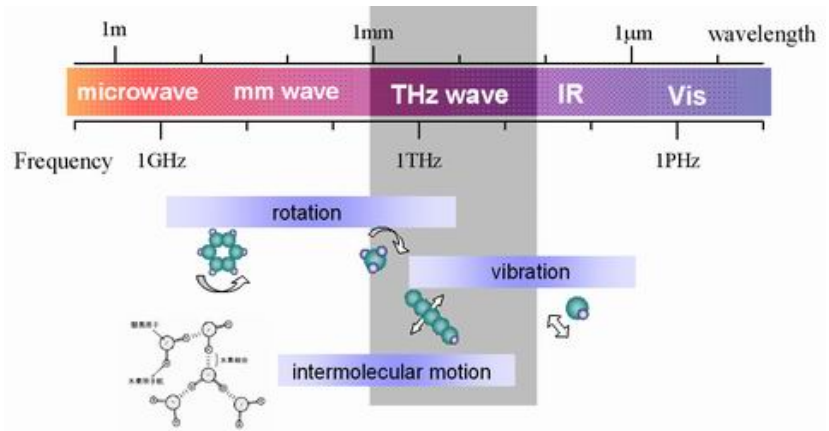
- black-body radiation
- bound-bound
  - electron transition atomic level
  - molecular transition (electronic, vibration, rotation)



Molecular spectroscopy

Various radiation processes

- black-body radiation
- bound-bound
  - electron transition atomic level
  - molecular transition (electronic, vibration, rotation)
- bound-free
  - ionization / recombination
- free-free
  - bremsstrahlung, cyclotron, synchrotron
- scattering processes
  - elastic scattering: Thomson, Rayleigh, Mie
  - inelastic scattering: Compton scattering (& inverse)
    - Sunyaev-Zel'dovich effect : distorsion of CMB by inverse Compton scattering

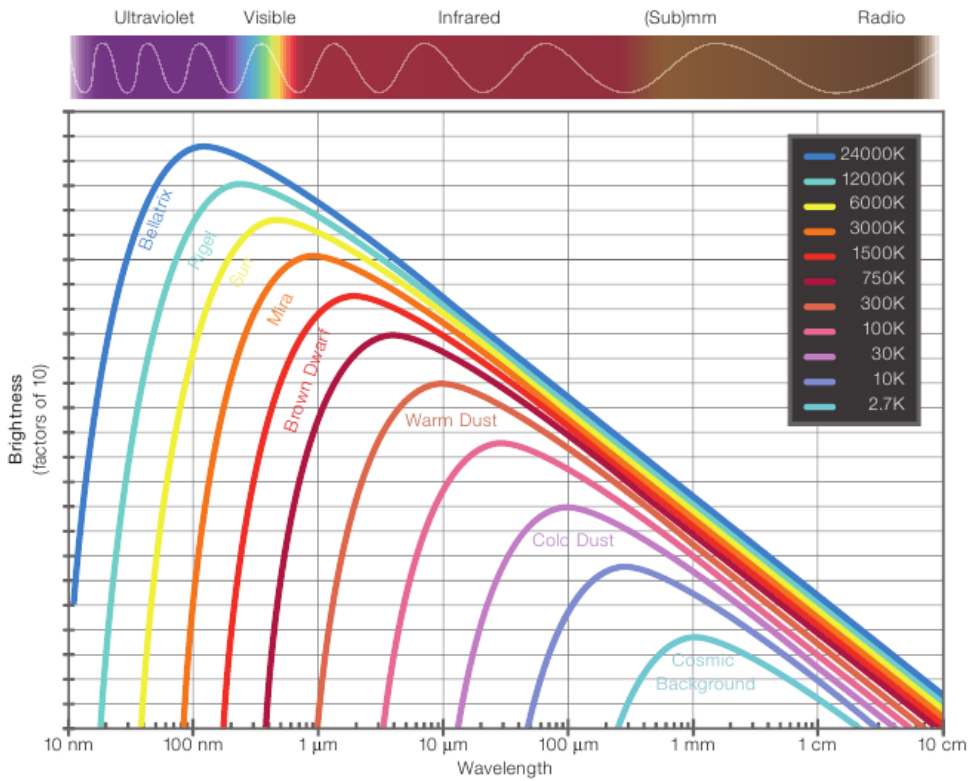


Molecular spectroscopy

Properties of the Black-Body Radiation (1/2)

- SED of an ideal radiator
  - Perfectly absorbing
  - Thermal equilibrium radiation ↔ matter
  - Pure continuum, no line
- Specific intensity

  intensité spécifique/luminance spectrale
- Unit $\mathrm{W.m^{-2}.Hz^{-1}.sr^{-1}}$
  - $B_\nu d\nu = \dfrac{2h\nu^3}{c^2} \dfrac{d\nu}{e^{\frac{h\nu}{k_B T}} - 1}$
  - $B_\lambda d\lambda = \dfrac{2hc^2}{\lambda^5} \dfrac{d\lambda}{e^{\frac{hc}{\lambda k_B T}} - 1}$
- Constants
  - $h = 6.626 \times 10^{-34} \mathrm{J \cdot s}$
  - $k_B = 1.381 \times 10^{-23} \mathrm{J \cdot K^{-1}}$

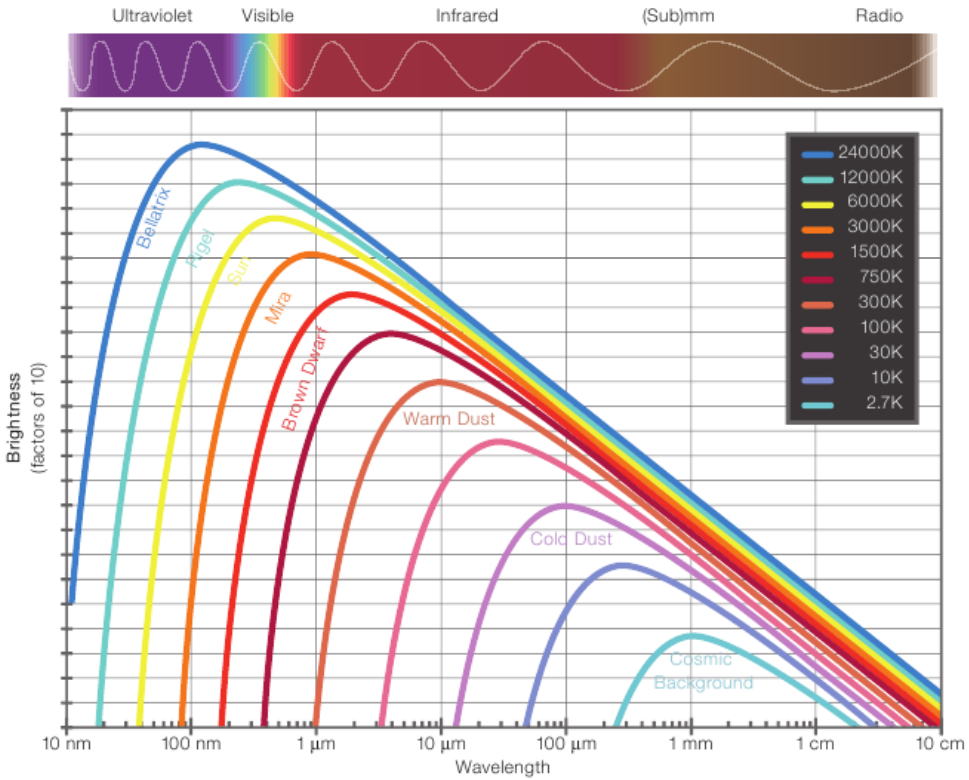


Blackbody SED as a function of temperature
Credit : Robert Hurt/ESO

Properties of the Black-Body Radiation (2/2)

- Stefan-Boltzmann's law
  - $F = \sigma T^4$, integrated flux at BB surface [W/m^2]
  - $\sigma = \dfrac{2\pi^5 k_B^4}{15 c^2 h^3} = 5.669 \times 10^{-8} \mathrm{W.m^{-2}.K^{-4}}$
- Wien's displacement law
  - $\lambda_{\max} = \dfrac{hc}{4.965 k_B T} = \dfrac{2\,898}{T}$ [μm]
- Rayleigh-Jeans law
  - For $\lambda \gg \dfrac{hc}{k_B T}$: $B_\lambda \simeq \dfrac{2ck_B T}{\lambda^4}$
- Wien approximation
  - For $\lambda \ll \dfrac{hc}{k_B T}$:

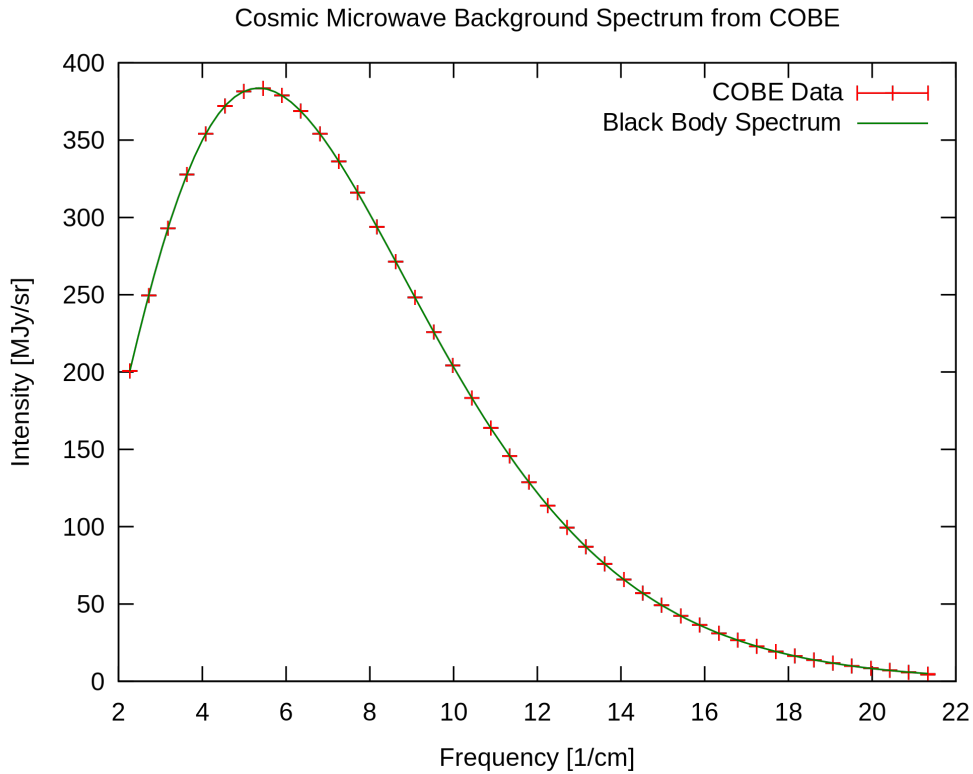
    $B_\lambda \simeq \dfrac{2hc^2}{\lambda^5} e^{-\frac{hc}{\lambda k_B T}}$



Blackbody SED as a function of temperature
Credit : Robert Hurt/ESO

Astrophysical black bodies

- Cosmic microwave background (CMB)

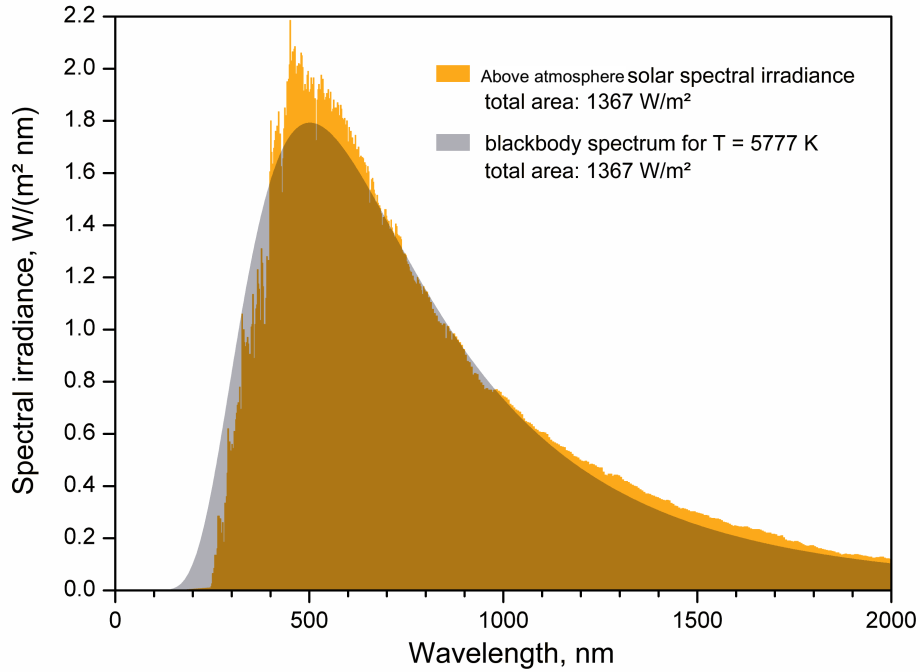


Blackbody SED fit to the CMB as observed by COBE/FIRAS

Credit : NASA

Astrophysical black bodies

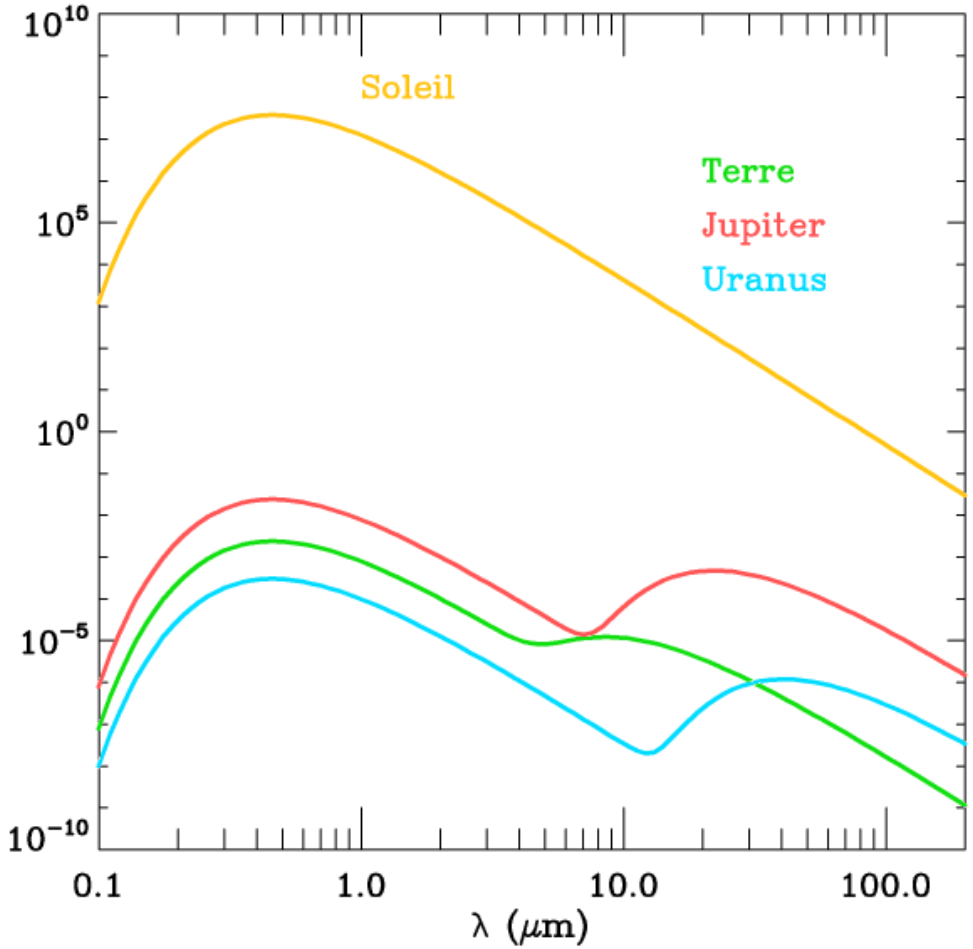
- Cosmic microwave background (CMB)
- Sun and most stars



Comparison between the Sun SED and a B-B with same $T_{\rm eff}$

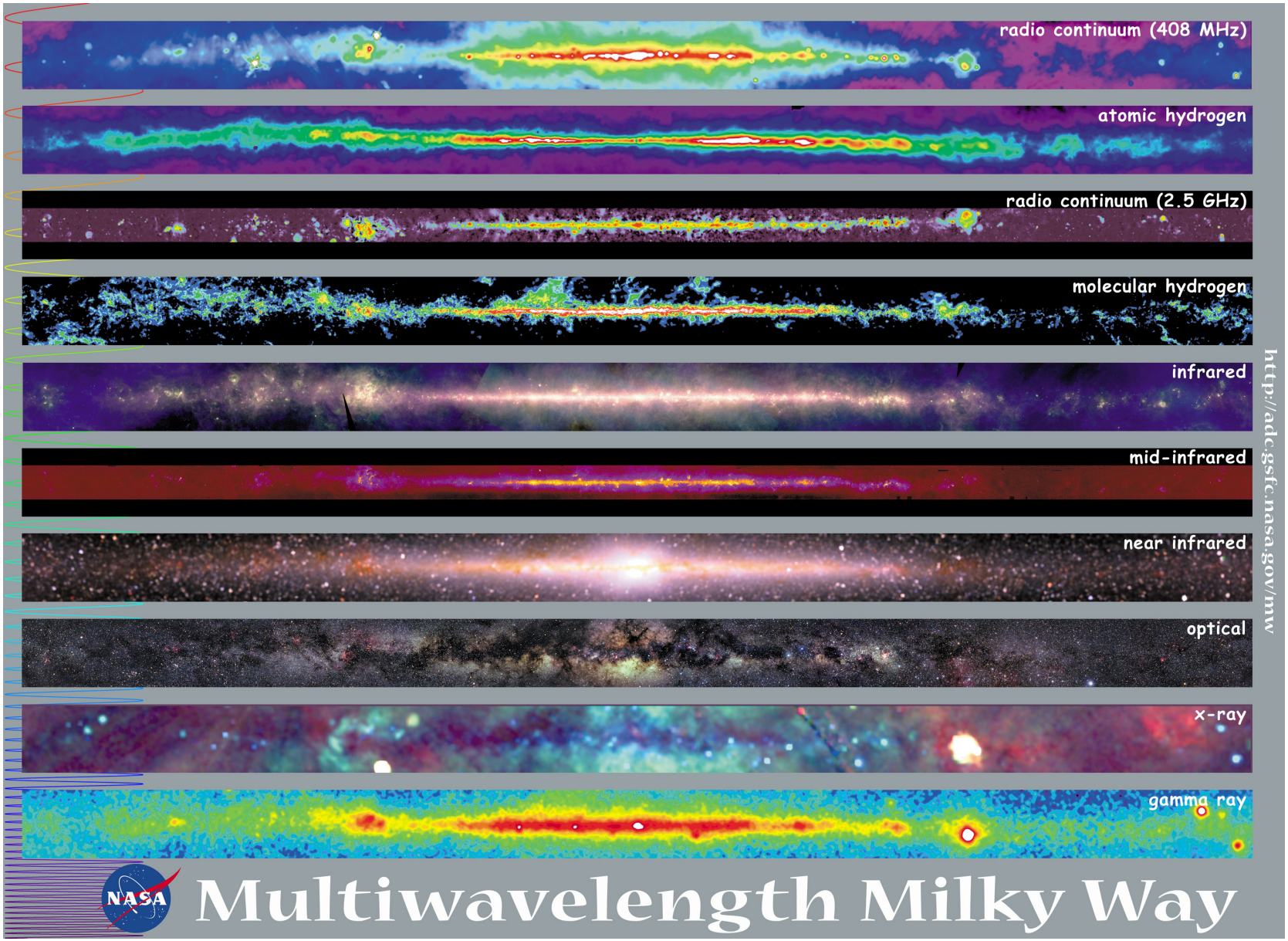
Astrophysical black bodies

- Cosmic microwave background (CMB)
- Sun and most stars
- Planets: composite spectrum



SEDs in the solar system

An example: the Milky-Way



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Astronomical Photometry : origins

- Hipparcos classified stars by brightness
  - 1 (brightest) to 6 (faintest)
- 1856: Pogson
  - m=6: 1/100 of the brightness of an m=1 star

➥ defines logarithmic scale based on $\sqrt[5]{100} = 2.512$

  - visual magnitude based on human eye
  - photographic magnitude based on photographic emulsions

Sun	-26.8
full Moon	-12.7
brightest Venus	-4.8
Sirius	-1.4
Vega	0.0
Polaris	2.0
Visual limit	6
HST deep field limit	30

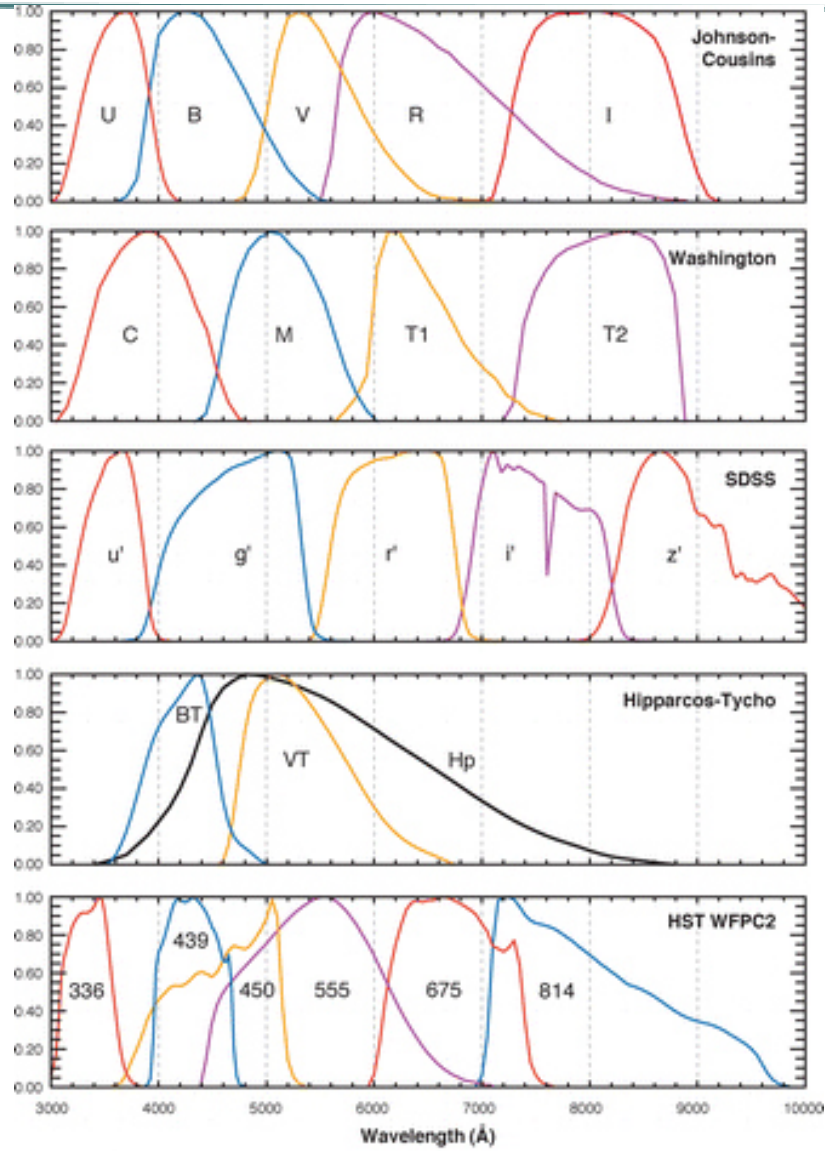
Astronomical Photometry : bolometric quantities

- Total energy radiated by a star :
  - $L_{\rm bol}$ (W)
- Flux collected at distance d from the star :
  - $F_{\rm bol} = \dfrac{L_{\rm bol}}{4\pi d^2}$ (W/m^2)
- Apparent bolometric magnitude :
  - $m_{\rm bol} = -2.5 \log_{10}\left(\dfrac{L_{\rm bol}}{4\pi d^2}\right) + C = -2.5 \log_{10}\left(\dfrac{F_{\rm bol}}{F_0}\right)$
  - The "zero" of the magnitude scale is defined by the constant C or equivalently by the constant F_0

Astronomical Photometry : filters

- Actual measurements are done through standardized filters
  - Visible : Johnson-Cousins system : U, B, V, R_c, I_c
  - Visible : Gunn/SDSS system : u, g, r, i, z
  - nIR : 2MASS : JHK_s
- Flux reference :
  - Vega system : m_F(Vega)=0.03 for any filter F (Johnson-Cousins)
  - AB : $m_{\mathrm{AB}} = -2.5 \log_{10} \left(\frac{F_\nu}{F_{\nu,0}} \right)$ with $F_{\nu,0} = 3.631 \times 10^{-23}\ \mathrm{W.m^{-2}.Hz^{-1}}$
- In practice...
  - comparison with “standard” stars
  - magnitudes are extrapolated above the atmosphere

Astronomical Photometry : filters



Bessell, MS. 2005
Annu. Rev. Astron. Astrophys. 43: 293–336

Astronomical Photometry : filters

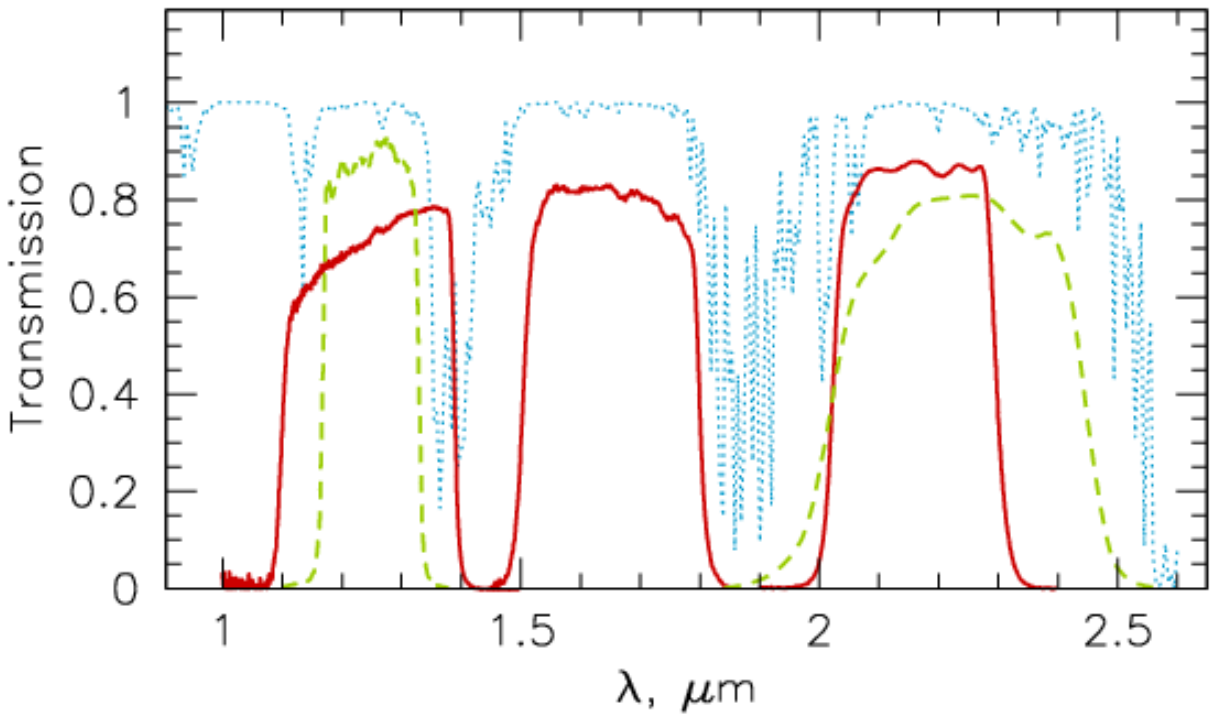


Figure: JHK Filter curves, and atmospheric transmission in the near-IR

Astronomical Photometry : filters

Photométrie standard

domaine spectral	Bande photométrique	λ (μm)	$\Delta\lambda$(μm)	Référence ($\mathrm{W\,m^{-2}\,\mu m^{-1}}$)
UV	U	0.36	0.068	$4.35\ 10^{-8}$
bleu	B	0.44	0.098	$7.20\ 10^{-8}$
visible	V	0.55	0.089	$3.92\ 10^{-8}$
rouge	R	0.70	0.22	$1.76\ 10^{-8}$
proche IR	I	0.90	0.24	$8.3\ 10^{-9}$
proche IR	J	1.25	0.30	$3.4\ 10^{-9}$
IR	H	1.65	0.35	$7.0\ 10^{-10}$
IR	K	2.20	0.40	$3.9\ 10^{-10}$
IR	L	3.40	0.55	$8.1\ 10^{-11}$
IR	M	5.0	0.3	$2.2\ 10^{-11}$

Astronomical photometry: absolute magnitude

- Absolute magnitude: apparent magnitude as measured at $d = 10$ pc
- The distance modulus is defined assuming that the line-of-sight is absorption-free :
  - $\mu = m_{\rm bol} - M_{\rm bol} = 5 \log_{10} \left(\frac{d}{d_0} \right) - 5$, where $d_0 = 1$ pc
- If interstellar absorption is not negligible:
  - $m = M + 5 \log_{10} \left(\frac{d}{d_0} \right) - 5 + A$
- Reference numerical values :
  - $M_{\rm bol} = 0 \Leftrightarrow L = 3.0128 \times 10^{28}$ W IAU 2015 B2 [arXiv:1510.06262]
  - $M_{\rm bol,\odot} = 4.74 \Leftrightarrow L_\odot = 3.828 \times 10^{26}$ W

Astronomical Photometry : useful relations

- Flux ratio from magnitude difference:
  - $\frac{F_2}{F_1} = 10^{-0.4(m_2 - m_1)}$
- Two stars w/ same intrinsic luminosity (valid for $m_{\rm bol}$):
  - $m_1 - m_2 = 5 \log_{10} \left(\frac{d_1}{d_2} \right)$
- Uncertainties and error propagation
  - $\delta \ln F = \frac{\ln 10}{2.5} \delta m$, with: $\frac{\ln 10}{2.5} = 0.92103$
- Usual non-SI units
  - $1\ \mathrm{Jy} = 10^{-26}\ \mathrm{W.m^{-2}.Hz^{-1}} = 10^{-23}\ \mathrm{erg.s^{-1}.cm^{-2}.Hz^{-1}}$

Astronomical Photometry : color indices

- Color or color index: mag difference in two filters of increasing wavelength
  - $B - V = m_B - m_V$

➔ independent of d but modified by interstellar extinction (absorption)

- Colors in Vega system: by def. Vega has all colors equal to 0
  - $B - V > 0$: redder than Vega
  - $B - V < 0$: bluer than Vega
- Bolometric correction: $BC = m_{\rm bol} - m_V$

Table 3.6 *Empirical Main Sequence*

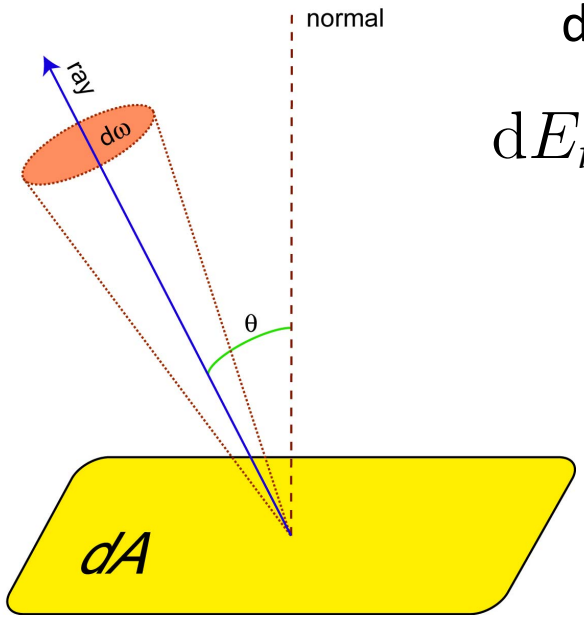
Spectral type	$B-V$	log $M\,M_\odot$	log $R\,R_\odot$	log T_e K	M_V	$M_{\rm bol}$	BC
O5	−0.34	1.4	0.9	4.60	−5.5	−8.1	−2.6
B0	−0.30	1.15	0.76	4.475	−4.0	−6.2	−2.2
B5	−0.17	0.64	0.505	4.190	−0.9	−2.1	−1.2
A0	0.00	0.324	0.262	3.974	1.5	1.3	−0.2
A5	0.13	0.222	0.185	3.911	2.30	2.33	0.03
F0	0.35	0.122	0.117	3.844	3.33	3.33	0.00
F5	0.45	0.067	0.072	3.810	3.91	3.90	−0.01
G0	0.61	0.006	0.018	3.772	4.63	4.57	−0.06
G5	0.66	−0.015	−0.026	3.762	4.85	4.75	−0.10
K0	0.85	−0.102	−0.083	3.715	5.83	5.63	−0.20
K5	1.20	−0.264	−0.227	3.633	7.60	7.19	−0.41
M0	1.4	−0.366	−0.339	3.604	9.50	8.06	−1.4
M5	1.6	−0.57	−0.54	3.513	13.0	9.98	−3.0
M8	1.8	−0.95	−0.96	3.418	19.5	12.93	−6.6

- Interstellar reddening
  - $A_V = 3.1\,E_{B-V}$ is representative for ISM ($E_{B-V} = A_B - A_V$)

Photometric quantities in astronomy

I_ν	specific intensity	intensité spécifique radiance	$\mathrm{W.m^{-2}.Hz^{-1}sr^{-1}}$
F_ν	monochromatic flux spectral irradiance	flux monochromatique (source) irradiance spectrale (recepteur) éclairement monochromatique	$\mathrm{W.m^{-2}.Hz^{-1}}$
F	flux irradiance (integrated in band)	flux (source) irradiance (recepteur) éclairement	$\mathrm{W.m^{-2}}$
F_{bol}	bolometric flux bolometric irradiance (whole spectrum)	flux bolométrique (source) irradiance bolométrique (recepteur) éclairement bolométrique	$\mathrm{W.m^{-2}}$
L_{bol}	bolometric luminosity	luminosité bolométrique	W

Specific intensity



Energy transported through a surface area dA, during dt, in frequency interval dν, and solid angle dΩ

$$\begin{aligned} \mathrm{d}E_\nu &\equiv I_\nu(\vec{r},\vec{l},t)\,(\vec{l}\cdot\vec{n})\,\mathrm{d}A\,\mathrm{d}t\,\mathrm{d}\nu\,\mathrm{d}\Omega \\ &= I_\nu(x,y,z,\theta,\varphi,t)\,\cos\theta\,\mathrm{d}A\,\mathrm{d}t\,\mathrm{d}\nu\,\mathrm{d}\Omega, \end{aligned}$$

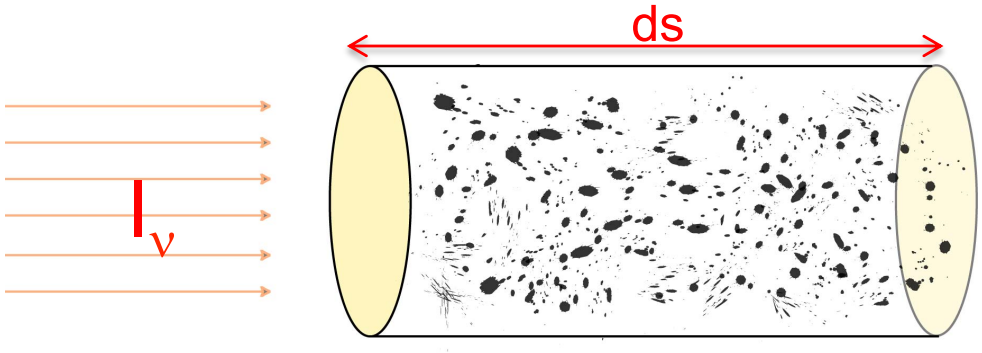
I_ν is the specific intensity (W/m^2/Hz/ster)

Note: $I_\lambda = I_\nu\, c/\lambda^2$ (W/m^3/ster)

$I \equiv \int_0^\infty I_\nu\,\mathrm{d}\nu$ is the total intensity

In the absence of absorption or emission, the specific intensity is conserved along a ray (energy conservation)

Interaction between photons and matter



Monochromatic **extinction coefficient**

per particle, or cross-section per particle (m^2):

$$\mathrm{d}I_\nu \equiv -\sigma_\nu n\, I_\nu\, \mathrm{d}s$$

per unit path length (m^{-1}):

$$\mathrm{d}I_\nu \equiv -\alpha_\nu\, I_\nu\, \mathrm{d}s$$

per unit mass, often called **opacity**, (m^2/kg), ρ being the density:

$$\mathrm{d}I_\nu \equiv -\kappa_\nu \rho\, I_\nu\, \mathrm{d}s$$

Optical thickness

If there is only extinction (no emission), over a distance D:

$$I_\nu(D) = I_\nu(0)\,\mathrm{e}^{-\tau_\nu(D)}$$

defining the **dimensionless optical thickness**:

$$\mathrm{d}\tau_\nu(s) \equiv \alpha_\nu(s)\,\mathrm{d}s; \qquad \tau_\nu(D) = \int_0^D \alpha_\nu(s)\,\mathrm{d}s,$$

An optical thickness of 1 implies $I_\nu = \frac{I_\nu^o}{e} \simeq 0.37\, I_\nu^o$

A medium is said to be optically thick/thin if $\tau > 1$ / < 1.

The optical thickness is related to the photon mean free path.

Radiative flux (1)

Net flux of energy through a surface dA (W/m^2/Hz)

$$\mathcal{F}_\nu(\vec{r},\vec{n},t) \equiv \int I_\nu \cos\theta \, \mathrm{d}\Omega = \int_0^{2\pi}\int_0^{\pi} I_\nu \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi.$$

In stellar atmospheres, customary to separate inward and outward flux (both positive):

$$\begin{aligned}
\mathcal{F}_\nu(z) &= \int_0^{2\pi}\int_0^{\pi/2} I_\nu \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi + \int_0^{2\pi}\int_{\pi/2}^{\pi} I_\nu \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi \\
&= \int_0^{2\pi}\int_0^{\pi/2} I_\nu \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi - \int_0^{2\pi}\int_0^{\pi/2} I_\nu(\pi-\theta) \cos\theta \sin\theta \, \mathrm{d}\theta \, \mathrm{d}\varphi \\
&\equiv \mathcal{F}_\nu^+(z) - \mathcal{F}_\nu^-(z),
\end{aligned}$$

For isotropic radiation: $\mathcal{F}_\nu^+ = \mathcal{F}_\nu^- = \pi I_\nu$ and $\mathcal{F}_\nu = 0$.

Axial symmetry:

$$\begin{aligned}
\mathcal{F}_\nu(z) &= 2\pi \int_0^{\pi} I_\nu \cos\theta \sin\theta \, \mathrm{d}\theta \\
&= 2\pi \int_0^{1} \mu I_\nu \, \mathrm{d}\mu - 2\pi \int_0^{-1} \mu I_\nu \, \mathrm{d}\mu \\
&= \mathcal{F}_\nu^+(z) - \mathcal{F}_\nu^-(z).
\end{aligned}$$

Notation: $\mu = \cos(\theta)$

Radiative flux (2)

Surface flux of a non-irradiated star: $\mathcal{F}_\nu^{\mathrm{surface}} \equiv \mathcal{F}_\nu^{+}(r=R)$

Flux received by observer at Earth (irradiance) from a star of radius R at distance D is :

$$f_\nu = \frac{4\pi R^2}{4\pi D^2}\,\mathcal{F}_\nu^{\mathrm{surface}}$$

Total flux: $\int_0^\infty \mathcal{F}_\nu \, d\nu = \mathcal{F}_{rad}$ (W/m²)

And stellar luminosity: $L = 4\pi R^2 \mathcal{F}_{rad}$ (W)

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Bibliographic ressources

- *"Astronomie Astrophysique, Introduction, 5e édition"*, A. Acker, 2013, Dunod
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